Enzyme Action: Preparing the pH Investigation

LAUNCH · Science · 40 minutes

I can explain enzyme specificity and denaturation, then identify the variables and evidence needed to investigate the effect of pH on amylase safely.

Prepare before pupils arrive

Setup: Provide the centre's current Pearson-aligned CP 1.10 method, risk assessment and labelled planning cards. Confirm reagents, concentrations, temperature control, timing, disposal and access arrangements with the technician before teaching.

Safety: Do not use this HTML as the practical method or risk assessment. No reagent or heating equipment is needed today. Correct the language 'enzyme killed' to 'enzyme denatured' without implying all pH effects are necessarily irreversible.

Equipment: Planning sheet, apparatus cards, pH-buffer labels, results-table template and current centre method/risk assessment supplied by the teacher. No reagent handling is required in this Introduce lesson.

Safety: This lesson plans but does not authorise practical work. The teacher/technician must supply the current centre-approved Pearson method, COSHH information and local risk assessment; pupils must not infer concentrations, heating arrangements or disposal from this deck.

Fallback: Sequence printed method cards and direct an adult to scribe exact decisions. Keep all GCSE variables and safety reasoning.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Edexcel GCSE Biology 1BI0 Foundation · Paper 1 · Topic 1 · enzyme action, pH and Core Practical 1.10 preparation.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 4 min	State what a biological catalyst does. Name the substrate broken down by amylase.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today’s answer.
Starter · 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask ‘What makes you think that?’ and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 5 min	specificity — one substrate fits denature — active site changes shape variable — what changes or is measured Represent amylase with one active-site shape and starch with a complementary substrate piece. Bind them briefly to show an enzyme-substrate complex.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 5 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	Translate the model into a testable question: how does buffer pH affect the time amylase takes to digest starch? Set the variables: IV = buffer pH; DV = time until a sample no longer turns iodine blue-black; CVs include temperature, enzyme and starch concentrations and volumes, sampling interval and mixing method.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Route each planning card to independent variable, dependent evidence or control/validity before writing the method. Read one planning card.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 10 min	Produce a safe, ordered CP 1.10 planning record that another trained person could follow using the centre-approved Pearson method and local risk assessment. Record: A pupil-owned prediction, IV/DV/CV table, ordered method outline, results-table headings and one justified safety/control decision.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Explain why temperature and volumes must be controlled when investigating pH and amylase. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Give one next move; the pupil edits or re-attempts on the existing work.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: An enzyme's active site has a complementary shape to its particular substrate.

Correct. Specificity depends on a complementary active-site and substrate relationship; the enzyme is not used up.

Misconception repair

Repaired. A valid comparison changes pH while controlling temperature, mixture volumes/concentrations, timing and sampling method.

Guided checkpoint

Correct. Specificity depends on a complementary active-site and substrate relationship; the enzyme is not used up. Return to the fit model. Which answer links one active-site shape to a particular substrate?

Buffer pH

INDEPENDENT VARIABLE. Independent variable: pH is deliberately changed. Ask which factor the investigation deliberately changes between mixtures.

Endpoint time

DEPENDENT EVIDENCE. Dependent evidence: endpoint time responds to the changed pH. Ask which observation is measured as the outcome.

Constant temperature

CONTROL / VALIDITY. Control variable: temperature must not become a second explanation for rate differences. Temperature is not the planned cause here; it must be controlled.

Equal volumes

CONTROL / VALIDITY. Control variables: equal volumes and concentrations make pH comparisons more valid. Ask whether this factor should differ between pH conditions.

Sampling interval

CONTROL / VALIDITY. Validity and precision: the same sampling rhythm supports comparable endpoint estimates. This is a method condition kept consistent, not the outcome itself.

No-amylase control

CONTROL / VALIDITY. Control sample: it checks that starch does not disappear without active enzyme. This sample is a comparison check, not a pH outcome used to calculate rate.

Model limitation

Card shapes do not show molecular movement, collisions, bonds or gradual changes in protein structure. A neat lock-and-key picture is a simplified representation, not a literal rigid lock.

Independent evidence

A pupil-owned prediction, IV/DV/CV table, ordered method outline, results-table headings and one justified safety/control decision.

Supported: Complete a labelled IV/DV/CV table and order six picture-method cards; finish one because sentence about keeping temperature constant. Use the word bank and sentence stem: We change __. We measure __. We keep __ the same because __.

Standard: Write an ordered method outline, identify variables and control sample, and design a table for pH, endpoint time and rate. Use numbered steps and the command-word frame: EXPLAIN = point + scientific reason.

Stretch: Justify the controls and sampling interval, predict the rate pattern, and identify one source of uncertainty before practical work begins. Distinguish validity, precision and safety; do not invent an unsupported concentration or heating method.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_L_W8L1_Enzyme_Action_Introduce.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.